Fiscal Year:	FY 2023	**Task Last Updated:**	FY 09/01/2022
PI Name:	Mourkioti, Foteini Ph.D.		
Project Title:	Telomere Length Regulation in Muscle Atrophy		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-6118	**Congressional District:**	3
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	11/01/2019	**End Date:**	09/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2023 per F. Hernandez/ARC (Ed., 2/27/23)		
Key Personnel Changes/Previous PI:	August 2021 report: The following changes in personnel occurred: NEW PERSONNEL: Erica Jones (Research Technician): Erica hired to replace James Kim and she works on the NASA project PERSONNEL CHANGE James Kim (Research Technician): James got accepted at the Master Program in Brown University. His last day in the lab was June 15th, 2022. Paola Kuri (postdoctoral fellow). Paola has finished her studies on January 2022. She moved to an industry career. Blake Hernandez (Research Specialist A/Technician). Blake received a fellowship to do a master's degree in Bioengineering. Harry Papaioannou (Penn undergraduate student). Harry finished his studies in the lab on February 2020. He is now on his last year of College of Arts and Science at UPenn. Jacob Kocan (Penn undergraduate student). Jacob graduated from College of Arts and Science in May 2021, and he got accepted at Med School at University of Pennsylvania, School of Medicine.		
COI Name (Institution):	Rompolas, Panteleimon Ph.D. (University of Pennsylvania)		
Grant/Contract No.:	80NSSC19K1602		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	(1) Introduction During the NASA Twins Study, telomere length changes were observed in blood cells. Telomeres are specific sequences at the end of the chromosomes, protecting them from deterioration, much like the plastic tip at the end of a shoelace. Muscle atrophy is one of the main characteristics of astronauts in space. The only way to minimize muscle atrophy is through intensive exercise every day. However, the effects of telomere length changes in adult muscle cells remain unexplored. Interestingly, our lab recently developed a new method which allowed, for the first time, telomere length measurements in specialized cells within the adult muscles, including human tissues. We found that telomere shortening is a distinct feature of muscle stem cells (MuSCs) in atrophied muscles as well as in stem cells after repeated cycles of muscle damage and repair, similar to the ones observed in intensive exercise. (2) Central objectives Aim 1: Determine the combined effects of weightlessness and radiation exposure on telomere length of MuSCs. Aim 2: Define the cellular outcomes MuSC-depended telomere length changes during muscle atrophy. Aim 3: Investigate the molecular mechanism of telomere length in MuSCs. (3) Methods to accomplish research objectives We will utilize the mouse hindlimb unloading model of disuse atrophy combined with gamma irradiation to evaluate their effect on telomere length, using a new method developed in our lab. Next, we will monitor for the first time in vivo the number of MuSCs using two-photon microscopy using genetically modified mice we recently generated and will further investigate the extent of apoptosis and senescence during atrophy. These studies are expected to provide a comprehensive analysis of the exact cellular dysfunction of shortened telomeres. Last, we will query the levels of a specific telomeric binding protein and its effect on persistent DNA damage at the telomeric sites. These experiments are expected to elucidate the molecular crosstalk leading to shortened telomeres in muscle cells. (4) Significance The goals of this research proposal align with NASA's emphasis on Animal biology (vertebrates) in Support of Human Space Exploration and the Cell and Molecular Biology guiding questions 1, 5, and 8 (to define the underlying mechanisms of the cells that are influenced by gravitational changes, causing tissue/organ dysfunction and affect stem cell function, respectively), as described at the Space Biology Science Plan 2016-2025. This work is highly relevant to the care of astronauts since we anticipate that our results will help understand the importance of telomere length alterations in stem cells during skeletal muscle atrophy that have not been studied before. Our study will benefit NASA and astronauts' health by 1) evaluating the combined effect of weightlessness and radiation exposure on telomere length in muscle atrophy and 2) assessing the molecular pathways affected during this process. Overall, by combining unique molecular and cellular biology approaches as well as genetically engineered mice that our lab has developed, we will determine the molecular mechanisms of telomere defects in MuSCs of atrophied muscles that can translate into benefits for NASA Exploration Mission. The regulatory mechanisms that we will explore in this proposal, using proven analogs such as mouse hindlimb unloading that induce biological effects similar to spaceflight, will be fundamental for future studies to investigate muscle specific telomere length changes in transitions from one gravity to another. Additionally, success in our experiments will inform similar mechanisms in musculoskeletal applications with stem cell dysfunction, such as muscle injuries after repeated exercises, providing benefits to the health and well being to both astronauts in space and those on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Success in our experiments will inform similar mechanisms in musculoskeletal applications with stem cell dysfunction, such as muscle injuries after repeated exercises, providing benefits to the health and well being to both astronauts in space and those on Earth.
Task Progress:	Shortened telomeres have been identified as a distinct feature of patients with severe muscle disease characterized by extensive muscle injury and repair (such as Duchenne muscular dystrophy). Using a protocol termed muscle quantitative fluorescence in situ hybridization (MuQ-FISH), developed earlier by the Mourkioti Lab, we were able to quantify telomere length and number in prospectively isolated muscle stem cells (MuSCs). We were also able to establish that telomere attrition is present in human dystrophic MuSCs, underscoring their importance in diseased regenerative failure. This grant aims to better understand the importance of telomere length alterations in MuSCs during skeletal muscle atrophy and determine the molecular mechanisms of telomere defects in MuSCs of atrophied muscles. The investigation has studied a specific pathway, NF-kB, which is tied to DNA transcription and inflammation response. The Mourkoti Lab created assays to examine internal signaling pathways in mice with dystrophy. We observed a high proportion of NF-kB activation in cells that had experienced chronic injury and repair. The elevated NF-kB response was found to coincide with the shortening of telomeres, which left the DNA ends exposed and susceptible to damage. Persistent high levels of NF-kB reduced a protein called Ku80. The reduction in Ku80 left telomeres unprotected, and stem cells were exhausted over time, meaning that muscles couldn't be repaired. NF-kB is the first genetic pathway found to directly affect telomere shortening. For the analysis of samples during this reporting period, we established and validated a modified version of the MuQ-FISH (Tichy and Mourkoti, 2021). We found that this method, which we called CRYO MuQ-FISH, was beneficial for measuring cryosections of murine skeletal muscle, leaving the MuSC niche intact. References: Tichy ED, Mourkioti F. "Telomere length assessments of muscle stem cells in rodent and human skeletal muscle sections." STAR Protocols. 2021 Dec 17;2(4):100830. Available online 14 September 2021. https:// ; PMID: 34585163; PMCID: PMC8450295
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Articles in Peer-reviewed Journals	Ma N, Chen D, Lee JH, Kuri P, Hernandez EB, Kocan J, Mahmood H, Tichy ED, Rompolas P, Mourkioti F. "Piezo1 regulates the regenerative capacity of skeletal muscles via orchestration of stem cell morphological states." Sci Adv. 2022 Mar 18;8(11):eabn0485. https://www.science.org/doi/10.1126/sciadv.abn0485 ; PMID: 35302846; PMCID: PMC8932657 , Mar-2022
Significant Media Coverage	Mourkioti F. "Pathway behind muscle breakdowns in Duchenne muscular dystrophy discovered." Penn Medicine Newsletter, May 11, 2021. https://www.pennmedicine.org/news/news-releases/2021/may/pathway-behind-muscle-breakdowns-in-duchenne-muscular-dystrophy-discovered , May-2021